

9. Hydrogen peroxide, H_2O_2 , is highly reactive and can cause damage to proteins and fatty acids in the human body.

(a) Hydrogen peroxide, H_2O_2 , can react to form hydroxyl radicals ($\text{HO}\cdot$) starting a chain reaction. AH, BH and CH represent molecules found in the human body.

Reaction step	Name of step
$\text{H}_2\text{O}_2 \rightarrow 2\text{HO}\cdot$	initiation
$\text{HO}\cdot + \text{AH} \rightarrow \text{HOH} + \text{A}\cdot$ $\text{A}\cdot + \text{BH} \rightarrow \text{AH} + \text{B}\cdot$ $\text{B}\cdot + \text{CH} \rightarrow \text{BH} + \text{C}\cdot$	
$\text{A}\cdot + \text{HO}\cdot \rightarrow \text{AOH}$	termination

(i) Complete the table to show the name of the missing step.

1

(An additional table, if required, can be found on page 35.)

(ii) The free radicals shown in the table are written as, for example, $\text{A}\cdot$

State what is represented by the symbol $\cdot$

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(b) When hydroxyl free radicals react with enzymes, they can break bonds between the protein molecule chains causing them to unfold and change shape.

State the term used to describe this change in shape.

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(c) Free radical scavengers are molecules that react to prevent chain reactions.

Give an example of a type of consumer product that has free radical scavengers added.

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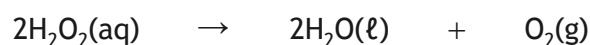
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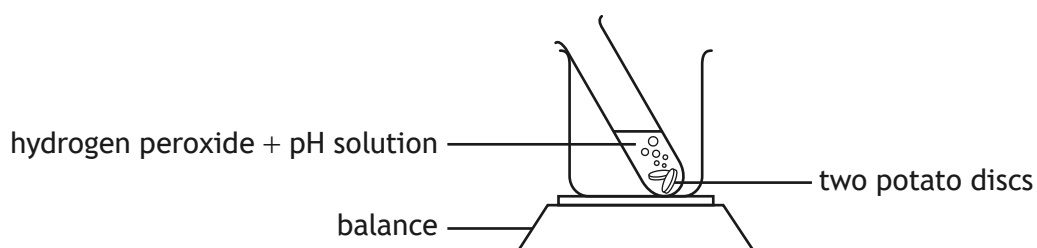
9. (continued)

- (d) In some living organisms an enzyme called catalase breaks down hydrogen peroxide into water and oxygen.

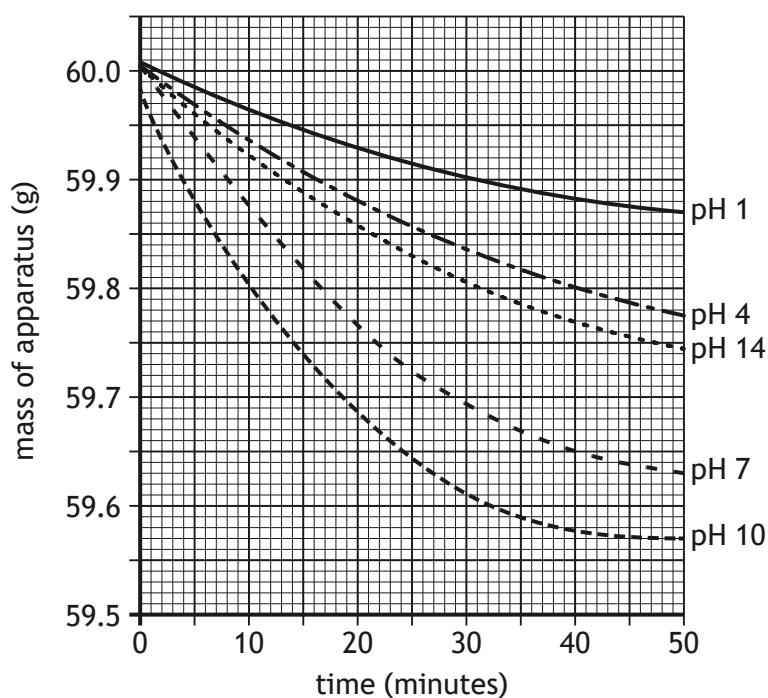


A student investigated how changes in pH affected the rate of reaction that was followed by measuring the mass lost over time.

Potato discs were placed in solutions with different pH values and the change in mass recorded.



The results are shown in the graph.



- (i) Suggest why the mass of the apparatus decreased over time.

1

9. (d) (continued)

- (ii) In each experiment, two potato discs were added to 1.0 cm³ of hydrogen peroxide and 15.0 cm³ of pH solution in a boiling tube.

State another **two** variables that the student should have kept constant.

2

- (iii) Using the graph, describe the effect that changing pH has on the rate of the breakdown of hydrogen peroxide by catalase in potatoes.

1

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